



UiT The Arctic University of Norway

Introduction of AI and ML Methods and Applications in Logistics Systems

Industry X.0 - INE-3612

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Introduction

Key Concepts of ML

Data Retrieval and Pre-processing

Algorithms and Techniques

Introduction

Stages of AI development

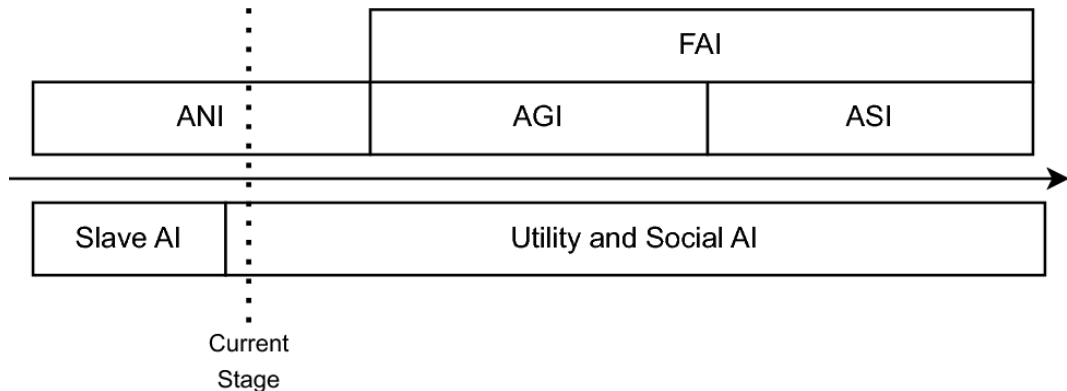


Figure 1: Stages of AI development from ANI, to AGI and ASI. Taken from Sun et al. (2026)

Logistics trends



Figure 2: Top ten logistics trends. Taken from maersk.com

What challenges does your company's logistics face?¹

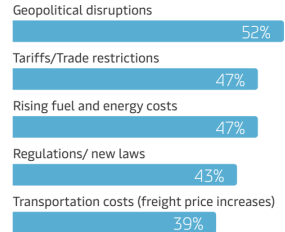


Figure 3: Logistics challenges. Taken from maersk.com

- Uncertainty increases complexity in logistics planning
- Customers expect for speed
- Raised concern in sustainability

Logistics trends (cont.)

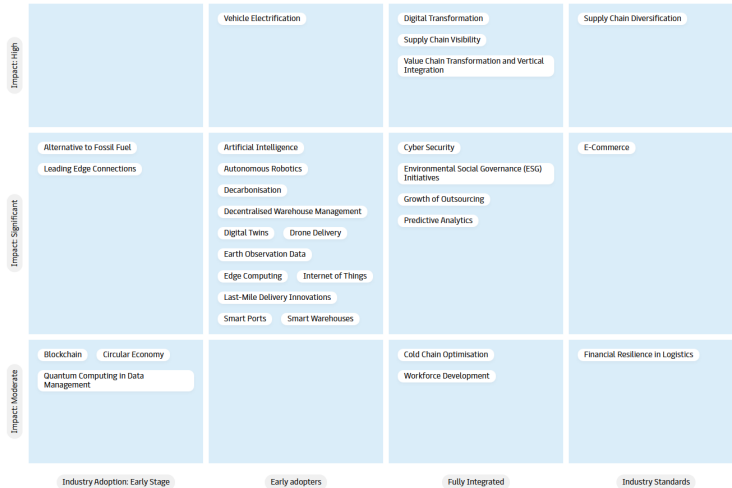


Figure 4: Logistics trend map. Taken from maersk.com

Logistics trends (cont.)

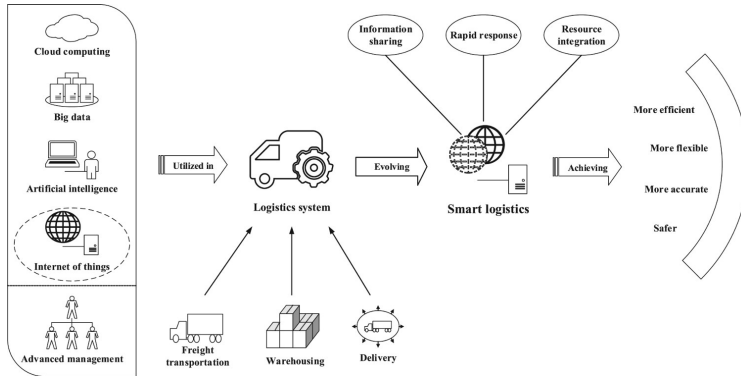


Figure 5: Smart logistics. Taken from Ding et al. (2021)

Types of logistics

Inbound logistics

Supplier → factory

Improves replenishment and reduces stockouts

Outbound logistics

Factory → warehouse or DC →
retailer or customer

Fulfill customer demand with high responsiveness

Reverse logistics

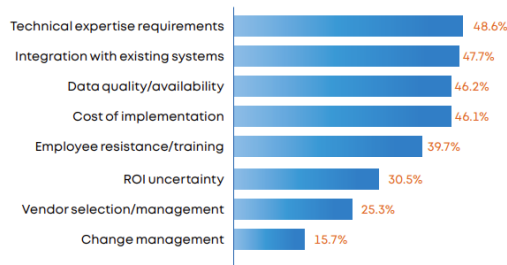
Return → repair → recycle

Defines customer's experiences to get returned value

Third-party logistics

- Transportation and fulfillment services by external partners
- Improves replenishment and reduces stockouts

- Demand planning
- Process optimization
- Routing planning
- Predictive maintenance of vehicle fleets
- Inventory management
- Warehouse management
-



● What are the main challenges you've faced in implementing AI/ML solutions?*

Figure 6: Main barriers to AI/ML adoption. Taken from ctl.mit.edu

Key Concepts of ML

Key Concepts of ML

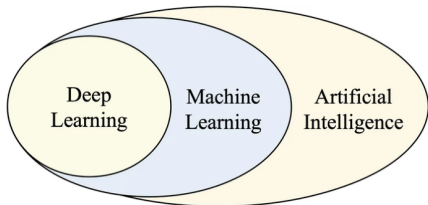


Figure 7: Subfields of AI. Taken from Wang (2025)

- **Data-driven**

- ▶ The goal is to extract valuable patterns from data
- ▶ Data quality plays a critical role in producing reasonable outputs

- **Model**

- ▶ It maps the input features to target outputs
- ▶ Good models generalize well to unseen data

- **Learning**

- ▶ Automatic pattern recognition
- ▶ Parameters optimization of the model

Types of ML

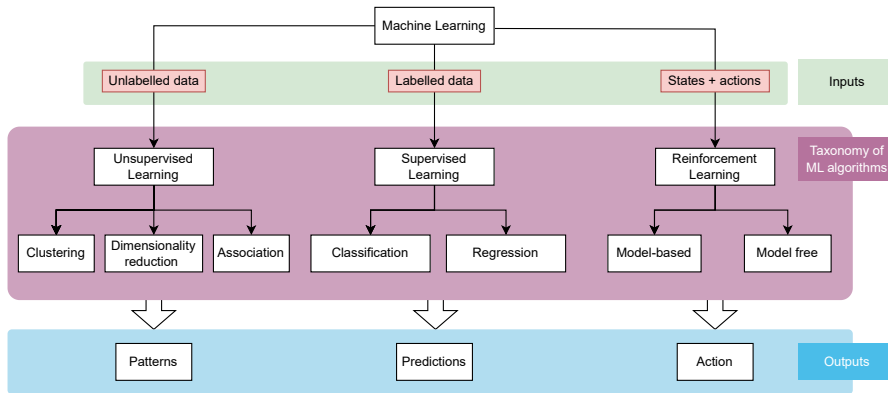


Figure 8: Taxonomy of ML algorithms

General Workflow

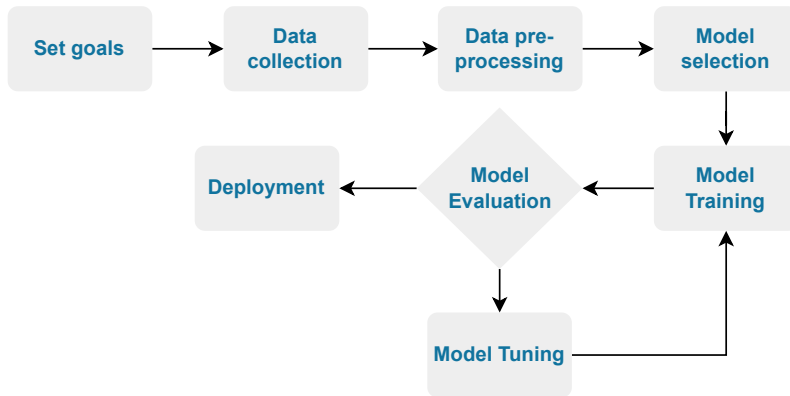


Figure 9: Typical ML workflow

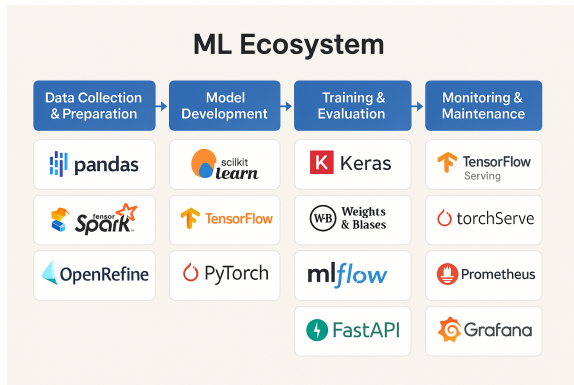


Figure 10: Popular ML tools. Generated by Copilot.

- GUI-based tools:
 - ▶ [RapidMiner](#)
 - ▶ [KNIME](#)
 - ▶ [AzureML](#)
- Interactive Development Environments (IDEs):
 - ▶ [Jupyter Notebook](#) hosted by UiT
 - ▶ [Google Colab](#)

Data Retrieval and Pre-processing

Table 1: Open data

Open data site	Description
Data.gov	The home of U.S. Government's open data
open-data.europa.eu	The home of European Commission's open data
Data.worldbank.org	Open data initiative from the World Bank
data.norge.no	Open data provided by Norwegian public sector
Opendatasoft	Market place of various open dataset
UCI ML repository	Collection of databases used by ML community
OpenML	Open platform for sharing datasets, algorithms, and experiments

Do data quality checks now! You will save your time and prevent further problems

Do data quality checks now! You will save your time and prevent further problems

- Cleansing
 - ▶ Missing values
 - ▶ Outliers
 - ▶ Spaces, typos, ...
- Integration
- Transformation
 - ▶ Aggregating
 - ▶ Extrapolating
 - ▶ Normalization

Pre-processing examples

Table 2.4 An overview of techniques to handle missing data

Technique	Advantage	Disadvantage
Omit the values	Easy to perform	You lose the information from an observation
Set value to null	Easy to perform	Not every modeling technique and/or implementation can handle null values
Impute a static value such as 0 or the mean	Easy to perform You don't lose information from the other variables in the observation	Can lead to false estimations from a model
Impute a value from an estimated or theoretical distribution	Does not disturb the model as much	Harder to execute You make data assumptions
Modeling the value (nondependent)	Does not disturb the model too much	Can lead to too much confidence in the model Can artificially raise dependence among the variables Harder to execute You make data assumptions

Figure 11: Handling missing data. Taken from Cielen and Meysman (2016).

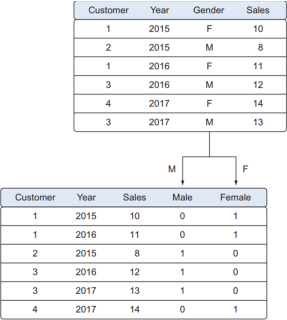


Figure 12: Transform data into dummies. Taken from Cielen and Meysman (2016).

Outliers removal

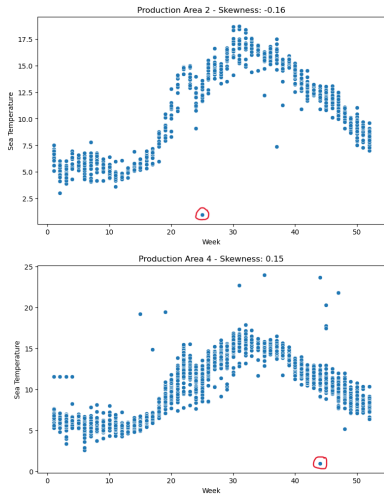


Figure 13: Outliers in sea temperature.

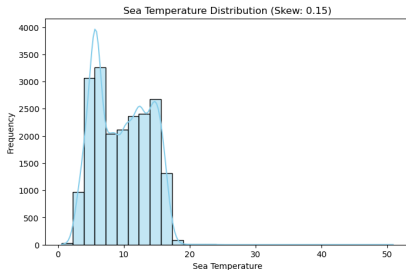


Figure 14: Data skewness of sea temperature.

- Z-Score
 - ▶ measures how many standard deviations a data point is from the mean
- IQR (Inter Quartile Range)
 - ▶ the range between Q1 and Q3 percentile
- Replace with mean/median

Outliers removal (cont.)

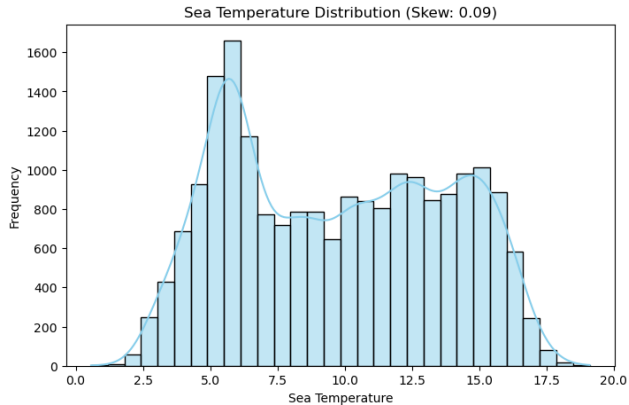


Figure 15: Data skewness reduce after outliers removal.

Transformation

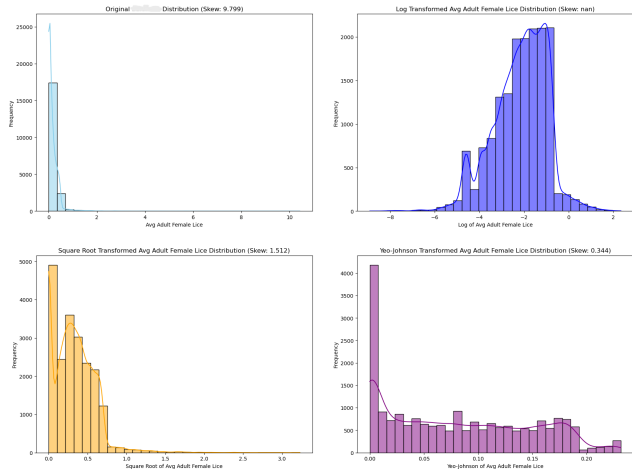


Figure 16: Transforming highly skew data

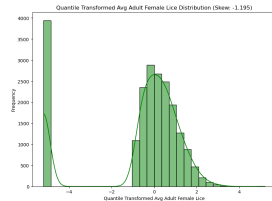


Figure 17: Quantile data transformation.

Transformation techniques:

- Min-max scaling
- Z-score scaling
- Log transformation
- Box-cox transformation
- Yeo-Johnson transformation
- Quantile transformation
- Square root transformation

Algorithms and Techniques



Figure 18: Unsupervised learning illustration

Table 2: Unsupervised learning algorithms.

Category	Algorithm	Use case example
Clustering	K-means clustering	Customer segmentation, supplier grouping by performance
	Dbscan clustering	Detecting anomalies in delivery patterns
	Hierarchical clustering	Analysing production batch similarities
Dimensionality reduction	PCA	Visualizing high-dimensional sensor data
Association mining	Apriori	Discover frequent items in order transaction
	FP-growth	Identifying co-occurring parts in inventory

Popular Algorithms (cont.)

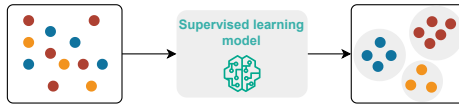


Figure 19: Supervised learning illustration

Table 3: Supervised learning algorithms.

Category	Algorithm	Use case example
Regression	Linear regression	Predicting maintenance cost
	Logistic regression	Predicting machine failure
Classification	KNN	Classifying defective and non-defective products
	SVM	Safety incidents classification
	Naïve Bayes	Transaction records filtering
	Neural Network (NN)	Product quality monitoring using image classification

Popular Algorithms (cont.)

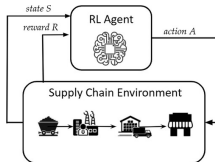


Figure 20: Reinforcement learning in SCM. Taken from Rolf et al. (2023)

Table 4: Refinforcement learning algorithms.

Category	Algorithm	Use case example
Model-free	Q-learning	Optimizing robot navigation, optimizing inventory
	Monte Carlo methods	Evaluating long-term supply chain policies
Model-based	Markov decision process (MDP)	Dynamic scheduling in manufacturing lines

- Validate the model against unseen dataset
- Standardize model's performance
- Measure model's ability to generalize the data
- Identify strategies (e.g., pre-processing, cross-validation, sampling) for improvement
- Measure choice depends on the nature of problem

Classification Evaluation Metrics

		Predicted	
		1	0
Actual	1	True Positive	False Negative
	0	False Positive	True Negative

Figure 21: Confusion matrix

Classification Evaluation Metrics

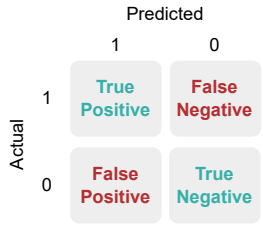


Figure 21: Confusion matrix

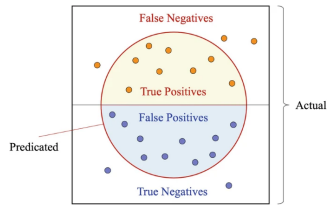


Figure 22: Illustration of classifier evaluation terms. Taken from Wang (2025).

Table 5: Classification metrics.

Metric	Formula	Intrepretation
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Overall correctness
Precision	$\frac{TP}{TP + FP}$	How reliable the positive prediction
Recall	$\frac{TP}{TP + FN}$	Ability to detect actual positive sample
Specificity	$\frac{TN}{TN + FP}$	Ability to detect actual negative sample
F1 score	$\frac{2TP}{2TP + FP + FN}$	Balance between precision and recall

What metric to choose?

- Measure choice depends on the nature of problem
- Consider the importance of identifying TP/TN and the cost of FP/FN
- **Recall** is used when missing a positive case is risky. Ex: predicting defective products using sensitive sensor
- **Specificity** ensures normal operations are not wrongly terminated. Ex: safety maintenance
- See the trade-off for classification metrics at [Precision & Recall](#)

Regression Evaluation Metrics

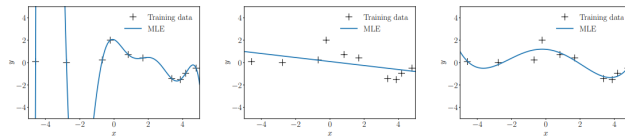
Table 6: Regression metrics.

Metric	Formula	Notes
Mean Absolute Error	$MAE = \frac{1}{n} \sum_{i=0}^{n-1} \ y_i - \hat{y}_i\ $	$\hat{y}_i$ is the predicted value of the i -th sample, and y_i is the corresponding true value. n is the number of samples
Mean Squared Error	$MSE = \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2$	
Mean Absolute Percentage Error	$MAPE = \frac{1}{n} \sum_{i=0}^{n-1} \frac{\ y_i - \hat{y}_i\ }{\max(\epsilon, \ y_i\)}$	ϵ is an arbitrary small yet strictly positive number to avoid undefined results when y is zero
Root Mean Squared Error	$RMSE = \sqrt{MSE}$	
R^2 Score	$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$	Represents the proportion of variance
Adjusted R^2	$AdjR^2 = 1 - (1 - R^2) \frac{n-1}{n-k-1}$	Comparing models with different numbers of features, often used in multiple regression. k is number of features or predictors

Clustering Evaluation Metrics

Metric	Notes
Silhouette Coefficient	Measure the similarity of a sample point in its own cluster vs others
Davies-Bouldin Index	Measure average similarity between clusters
Rand index	Similarity between predicted and true clusters, adjusted for chance

Model Fitting



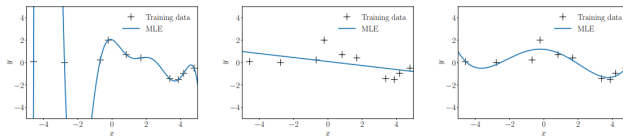
(a) Overfitting

(b) Underfitting.

(c) Fitting well.

Figure 23: Fitting problems in ML. Taken from Deisenroth et al. (2020).

Model Fitting



(a) Overfitting

(b) Underfitting.

(c) Fitting well.

Figure 23: Fitting problems in ML. Taken from Deisenroth et al. (2020).

Underfitting

- Build more complex model
- Add relevant features
- Feature engineering
- Check data quality
- Adjust training parameters (e.g., reduce regularization)

Overfitting

- Perform appropriate cross-validation
- Apply regularization
- Remove irrelevant features
- Simplify the model
- Consider early stopping

. : Continue to Part 2 ->

References

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